Integration — Lesson 21 Test: Completing the Power Rule

10 questions · show your work in the box · rewrite as powers of x first · don't forget + C on indefinite integrals

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Rewrite as a power of x, then integrate: $\int 1/x^3 \, dx$. Check your answer by differentiating. 1 pt

2. Integrate: $\int x^{1/4} \, dx$. (Remember: dividing by a fraction means multiplying by its flip.) 1 pt

3. Integrate: $\int 12/x^4 \, dx$. 1 pt

4. Integrate: $\int (\sqrt{x} + 4) \, dx$. 1 pt

5. Split the fraction, then integrate: $\int (x^4 + 5)/x^2\, dx$.

1 pt

6. Evaluate exactly: $\int_1^3 2/x^2\, dx$.

1 pt

7. Evaluate exactly: $\int_0^9 \sqrt{x}\, dx$.

1 pt

8. Evaluate exactly: $\int_4^9 1/\sqrt{x}\, dx$.

1 pt

9. A student writes $\int 1/x^2\,dx = \ln(x^2) + C$. Explain the error (when is ln the right tool, and when is the power rule?), and give the correct answer. 1 pt

10. Sap flows out of a maple tree at the rate $r(t) = 6/\sqrt{t}$ liters per hour, for t from 1 to 4 hours. (a) Rewrite r(t) as a power of t. (b) Find the total liters collected from $t = 1$ to $t = 4$. 1 pt